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CMSC 126

Entity: Smartphone

Data Dictionary

Data Element	Description	Data Type	Length/Size	Domain/Allowed Values
phone_id	Unique identifier assigned to each smartphone	INT	11	Positive Integer, Unique, Auto-Increment
brand	Manufacturer name of the device	VARCHAR	50	Alphabetic Characters
model_name	Specific model name of the device	VARCHAR	100	Alphanumeric Characters
color	Color variant of the device	VARCHAR	50	Alphabetic Characters
ram_gb	Random Access Memory size in GB	INT	11	Positive Integer
storage_gb	Internal storage capacity in GB	INT	11	Positive Integer
price	Retail selling price	DECIMAL	10,2	Positive Decimal Number
quantity	Number of units in stock	INT	11	Positive Integer

Database Design

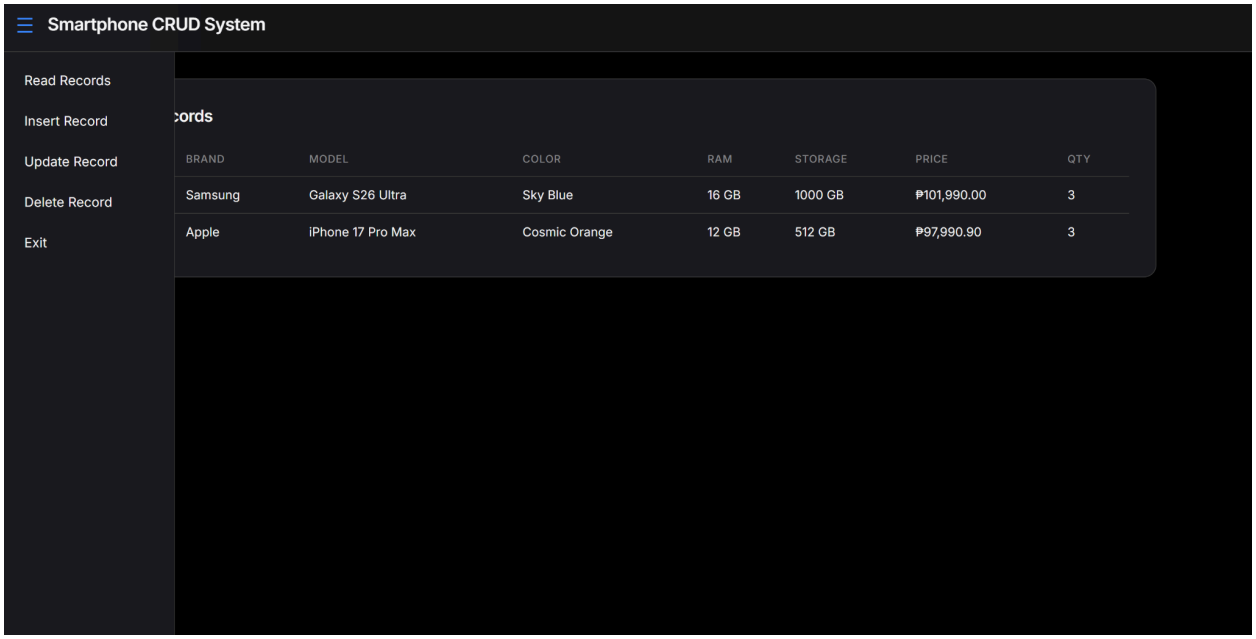
```sql
CREATE TABLE smartphones (
    phone_id INT AUTO_INCREMENT PRIMARY KEY,
    brand VARCHAR(50),
    model_name VARCHAR(100),
    color VARCHAR(50),
    ram_gb INT(11),
    storage_gb INT(11),
    price DECIMAL(10,2),
    quantity INT(11)
)
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0117 seconds.)

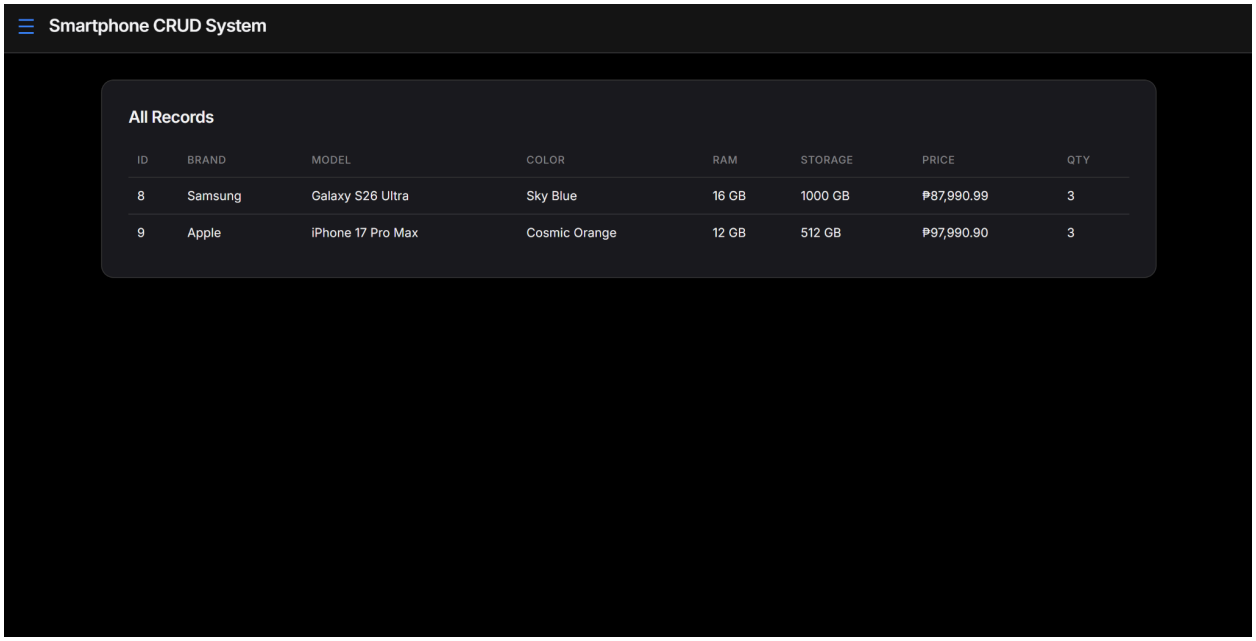
```sql
CREATE TABLE smartphones ( phone_id INT AUTO_INCREMENT PRIMARY KEY, brand VARCHAR(50), model_name VARCHAR(100), color VARCHAR(50), ram_gb INT(11), storage_gb INT(11), price DECIMAL(10,2), quantity INT(11) );
```

[Edit inline] [Edit] [Create PHP code]

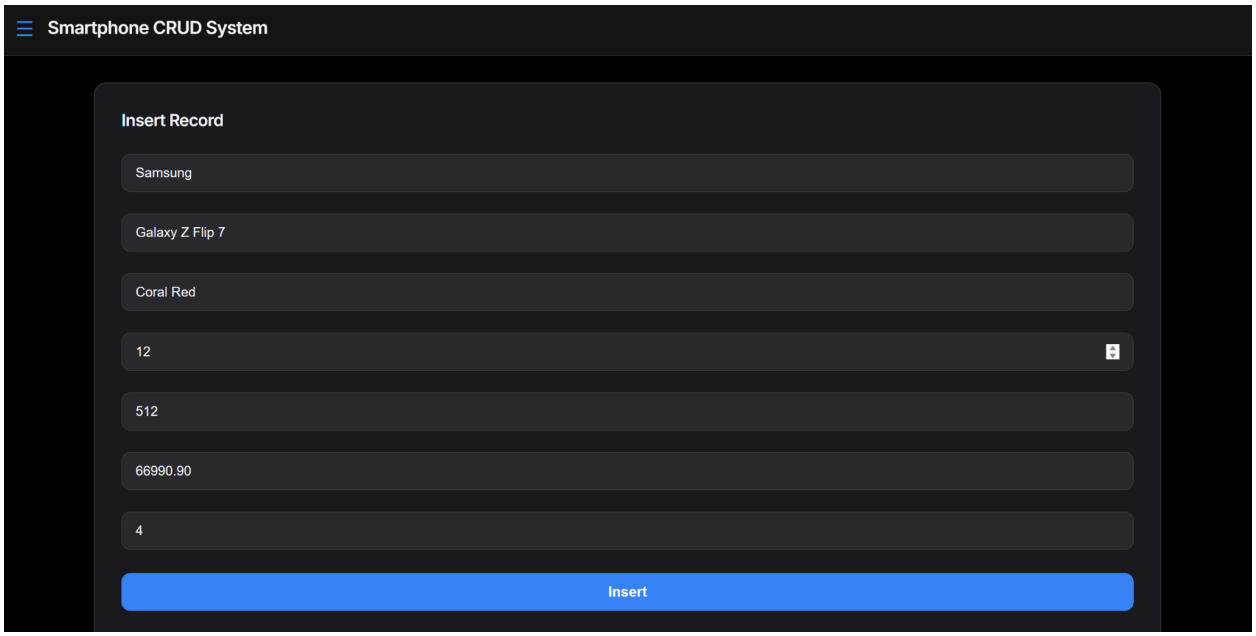
MENU INTERFACE

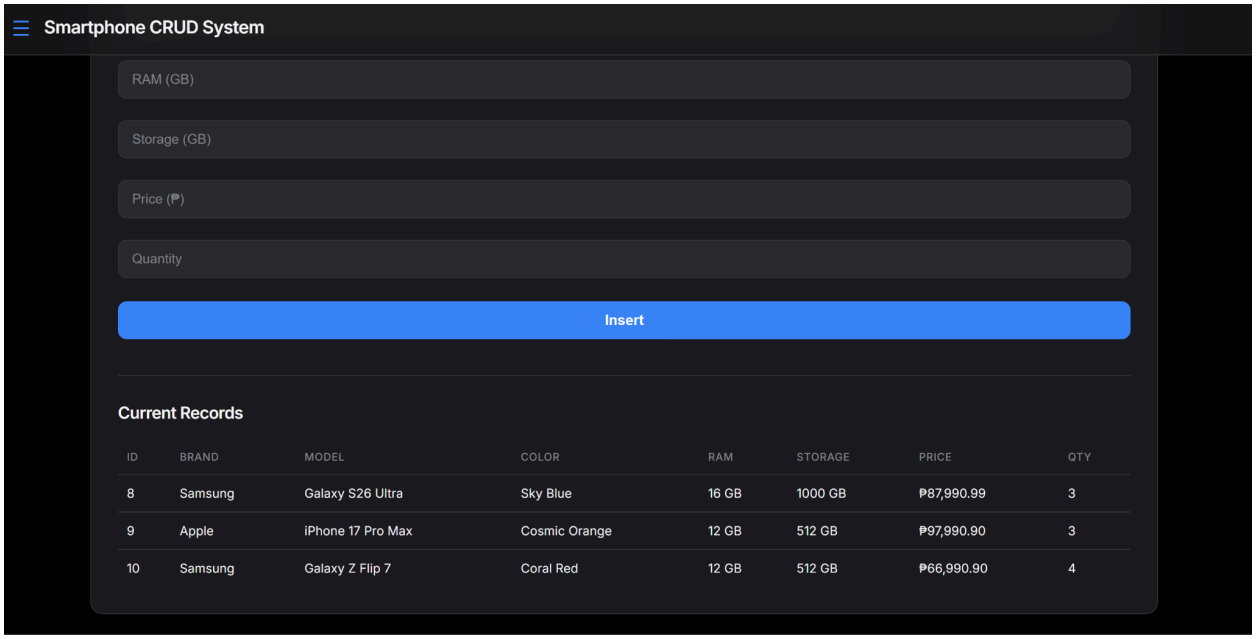


READ



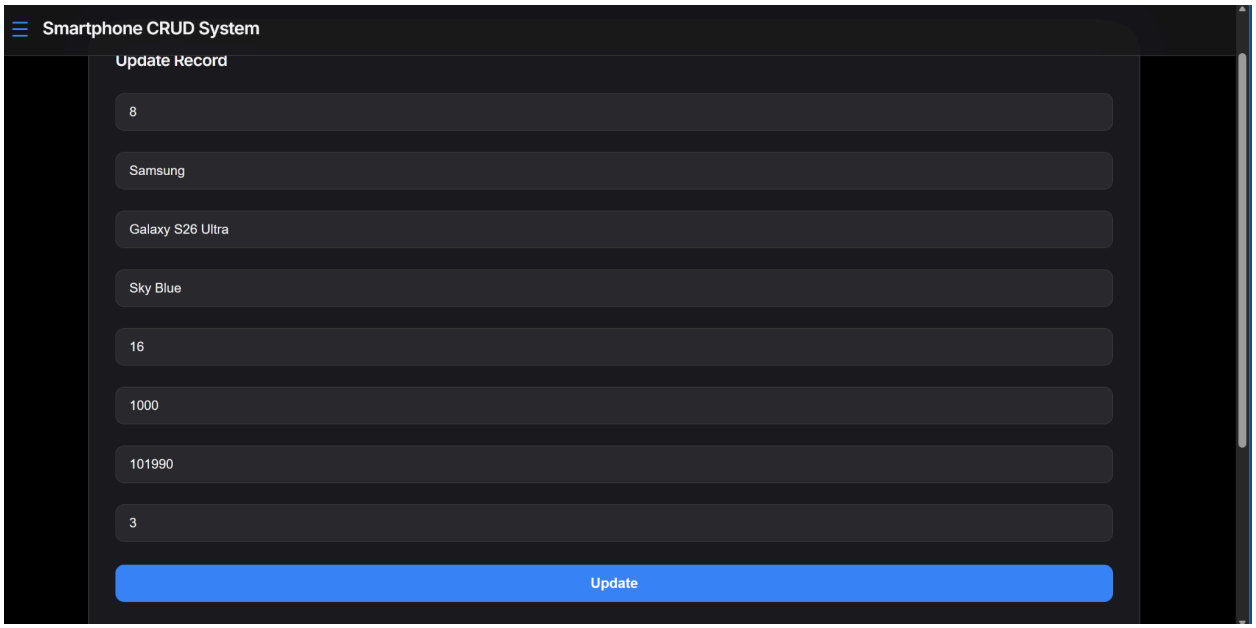
INSERT

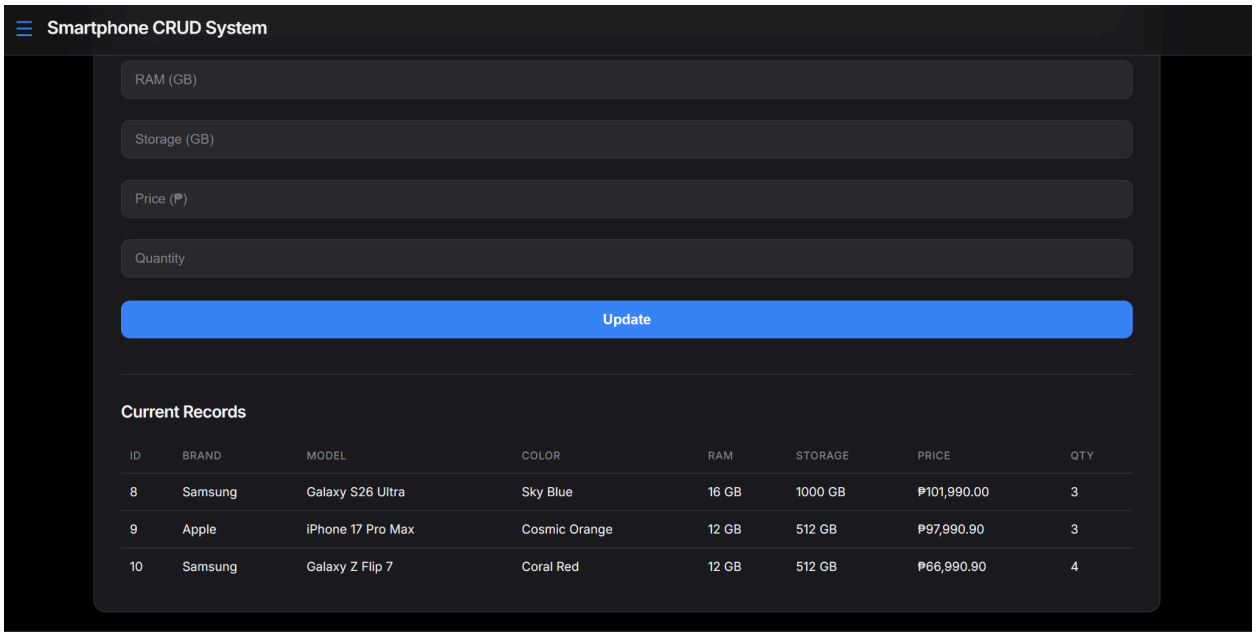




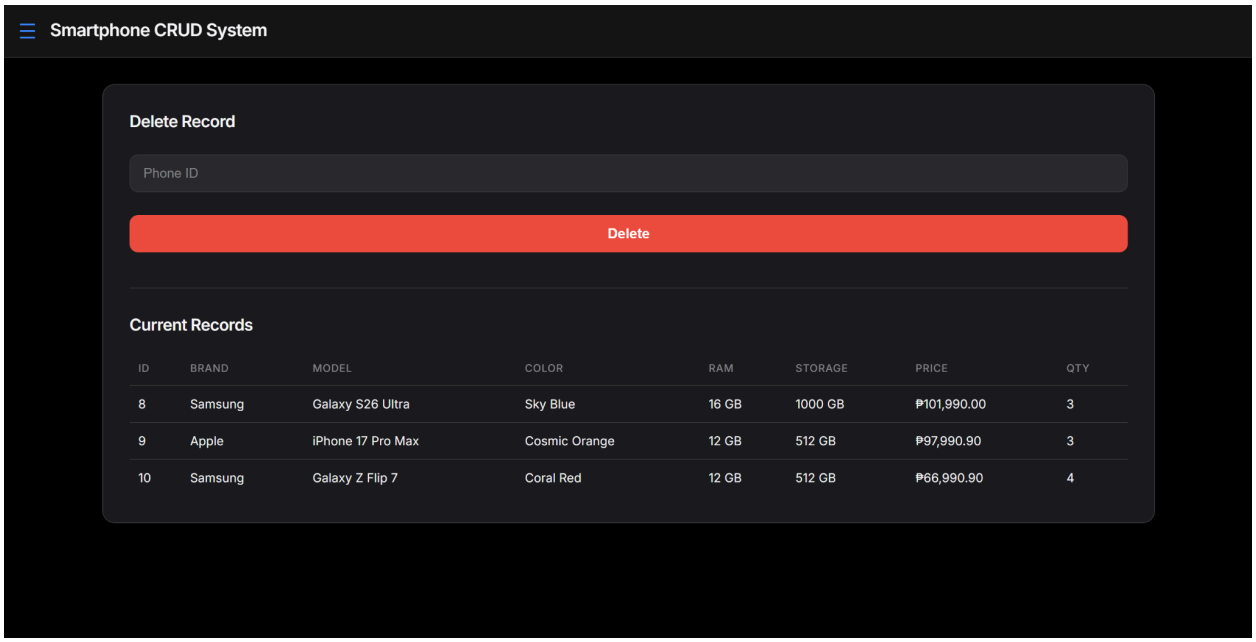
ID	BRAND	MODEL	COLOR	RAM	STORAGE	PRICE	QTY
8	Samsung	Galaxy S26 Ultra	Sky Blue	16 GB	1000 GB	₱87,990.99	3
9	Apple	iPhone 17 Pro Max	Cosmic Orange	12 GB	512 GB	₱97,990.90	3
10	Samsung	Galaxy Z Flip 7	Coral Red	12 GB	512 GB	₱66,990.90	4

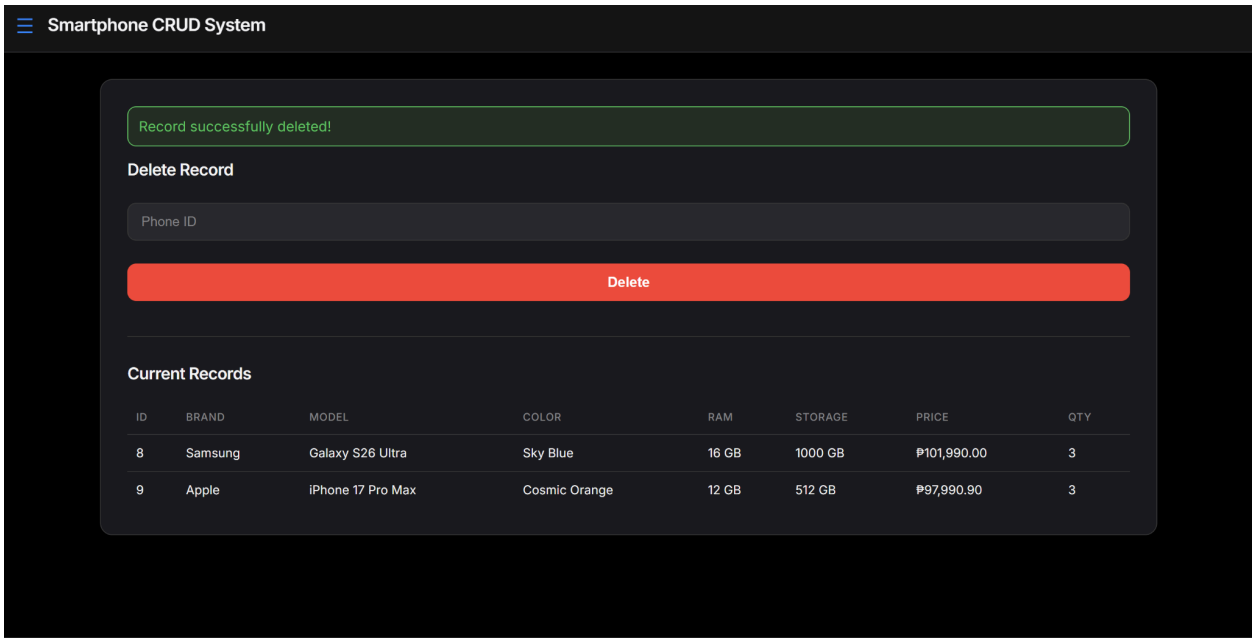
UPDATE


Smartphone CRUD System
Update Record
8
Samsung
Galaxy S26 Ultra
Sky Blue
16
1000
101990
3
Update

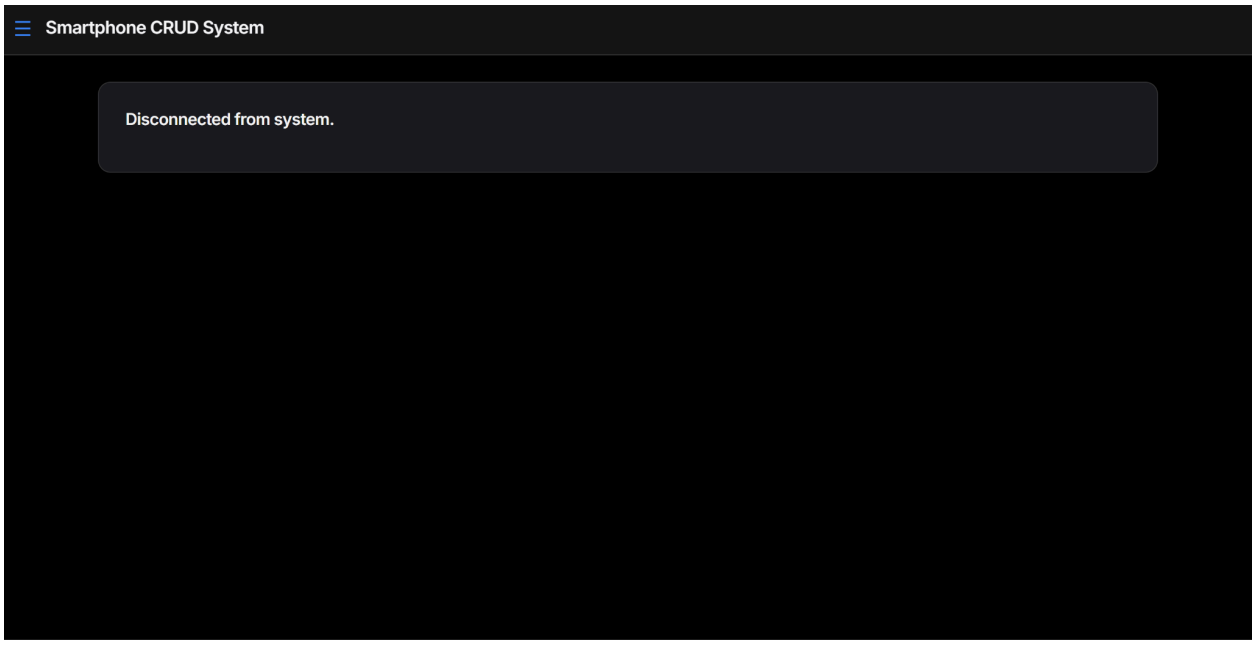

Smartphone CRUD System
RAM (GB)
Storage (GB)
Price (₱)
Quantity
Update
Current Records
ID BRAND MODEL COLOR RAM STORAGE PRICE QTY
8 Samsung Galaxy S26 Ultra Sky Blue 16 GB 1000 GB ₱101,990.00 3
9 Apple iPhone 17 Pro Max Cosmic Orange 12 GB 512 GB ₱97,990.90 3
10 Samsung Galaxy Z Flip 7 Coral Red 12 GB 512 GB ₱66,990.90 4

DELETE


Smartphone CRUD System
Delete Record
Phone ID
Delete
Current Records
ID BRAND MODEL COLOR RAM STORAGE PRICE QTY
8 Samsung Galaxy S26 Ultra Sky Blue 16 GB 1000 GB ₱101,990.00 3
9 Apple iPhone 17 Pro Max Cosmic Orange 12 GB 512 GB ₱97,990.90 3
10 Samsung Galaxy Z Flip 7 Coral Red 12 GB 512 GB ₱66,990.90 4


Smartphone CRUD System
Record successfully deleted!
Delete Record
Phone ID
Delete
Current Records
ID BRAND MODEL COLOR RAM STORAGE PRICE QTY
8 Samsung Galaxy S26 Ultra Sky Blue 16 GB 1000 GB ₱101,990.00 3
9 Apple iPhone 17 Pro Max Cosmic Orange 12 GB 512 GB ₱97,990.90 3

EXIT



1. What is the purpose of a data dictionary in database design?

A data dictionary acts as the ultimate "blueprint" or reference guide for your database. It contains metadata (data about data) that defines the structure, data types, and rules for every single table and column in your system.

If a new developer joins a project and looks at a table, they might not know what the rules are just by looking at the data. The data dictionary tells them exactly what to expect. For example, a data dictionary for your system would specify that `ram_gb` must always be an integer, `price` is a decimal value, and `brand` is a string with a maximum character limit.

2. Why must every table have a primary key?

A primary key guarantees entity integrity, meaning every single row in a table is uniquely identifiable. Without a primary key, it is impossible to reliably differentiate between two identical sets of data.

If you have two identical "Samsung Galaxy S24 Ultra" entries in your table with the exact same color, RAM, and price, a primary key like `phone_id` is the only way the database knows which specific one you are talking about. As you saw in your code, without that unique `phone_id`, trying to run an `UPDATE` or `DELETE` command could accidentally modify or wipe out every identical phone in the database at once.

3. What is the role of `isset()` in processing form input?

In PHP, `isset()` checks if a variable has been declared and is not null. When processing forms, its primary role is to verify that a user actually submitted the form before the server tries to run any database commands.

When a user simply loads your page, the PHP script runs from top to bottom. If you didn't wrap your insert logic inside `if (isset($_GET['submit_insert']))`, PHP would blindly attempt to run the `INSERT INTO` SQL query the moment the page opened, resulting in a blank, broken record being saved to your database every time someone visited the site.

4. How does a menu-driven interface improve usability?

Menu-driven interfaces drastically reduce a user's cognitive load. Instead of forcing the user to recall specific commands, syntax, or URLs from memory (like typing out raw SQL statements in a terminal), a menu presents all available options visually.

By using a clean sidebar with clear options like "Read Records" or "Insert Record," you change the user's task from *recall* (remembering what to do) to *recognition* (recognizing what they want to do from a list). This makes the software intuitive, faster to navigate, and highly accessible to non-technical users.

5. Why are CRUD operations fundamental in database systems?

CRUD (Create, Read, Update, Delete) represents the four absolute minimum operations required for any persistent storage system to be useful. If a system lacks even one of these capabilities, it becomes fundamentally incomplete or broken. Without the ability to create, you can never add new data, and without reading, you can never view or retrieve the information you have saved. Furthermore, without updating, your data becomes completely static and can never reflect changes in the real world, while lacking the ability to delete means your database will eventually become overwhelmed with outdated, irrelevant, or incorrect information.

REFERENCES

https://www.w3schools.com/howto/howto_js_mobile_navbar.asp

Used *Gemini AI* to enhance sentence structure and to help me debug.

Academic Integrity Oath

I certify that this laboratory work is my own original work. I did not copy any part of this program from other students or unauthorized sources. I understand that if this work is found to be copied or plagiarized, it may be subject to investigation and disciplinary action in accordance with university academic integrity policies.

Student Name & Signature: _____Charles O. Solomon_____
Date: __3/17/2026___